



TECHNICAL UNIVERSITY OF KENYA

FACULTY OF APPLIED SCIENCES AND TECHNOLOGY
SCHOOL OF COMPUTING AND INFORMATION TECHNOLOGY
DEPARTMENT OF INFORMATION SYSTEMS TECHNOLOGY
UNIVERSITY EXAMINATION SERIES
BACHELOR OF TECHNOLOGY COMPUTER TECHNOLOGY
BACHELOR OF TECHNOLOGY INFORMATION TECHNOLOGY
BACHELOR OF TECHNOLOGY COMPUTER COMMUNICATION
NETWORKS

ECII/ECSI/ECCI 4204: EMERGING TECHNOLOGICAL TRENDS IN ICT AND
SOCIETY

QUESTION 1 (30 Marks)

- a) Define the following terminologies and give one relevant example for each [3 marks]
- i) Emerging Technology
 - ii) Trend
 - iii) Disruption
- b) Some tech pundits believe that the current trend of foldable smartphones marks the beginning of the end of smartphones as we know it. They further predict a future where the era of the smartphone is over. In contrast, Google's newest idea is to stop us from spending more time on our phones. To achieve this goal Google has designed a special sealed envelope which you can place your phone in. Once your phone is inside the smart envelope only the phone camera and basic keypad to dial numbers can be used. Use the two contrasting definitions of emerging trends according to technology management authors Rotolo and Halaweh to determine which of the two trends between foldable smartphone and Google smart envelope will bring about a true paradigm shift for smartphones? [5 marks]
- c) Discuss the link between I.T. asset management software and patch management software in the context of their application in simplifying management of multi-user computing environments such as a computer lab [4 marks]
- d) Describe the future projected impact on i) society and ii) business within the next five-ten years for the following technology:-
- (i) IBM Food Trust's block chain for advanced food tracking and packaging [3 Marks] (e) Differentiate between the current telephone centric data communication networks and the future all I.P. (internet protocol) based next generation data networks [2 marks]
- f) Explain the relationship between Nielsen's law of internet bandwidth and computing power [4 marks]
- g) According to a popular American media technology website cnet.com, the following trends from the past decade 2010-2019 were either a failure or successes. For each technology provide an argument why it may not necessarily have succeeded or failed despite cnet's opinion [6

marks]

1. Non-traditional keyboard centric computers

2. Libra cryptocurrency

h) The use of in-store robots as customer service assistants in popular retail shops such as Carrefour Kenya is related to the current on-going 4th industrial revolution business and commerce innovations worldwide. Justify why their usage will significantly impact society in a different way compared to previously before they were not in use [3 marks]

QUESTION TWO (20 Marks)

a.) You have recently graduated after successfully completing your computer science qualification. Technical university of Kenya hires you to convert room D31 in Block D school of computing into a computer lab. They also want you to manage this new lab before handing it over back to them. Develop a detailed computer management plan showing what tasks, actors and roles you would require to install and run 10 machine hardwares, with software and a network in this new lab. Use best practices and emerging trends in your plan, and justify what type of computer lab designs and layouts you would suggest for them to implement and why [6 Marks]

b.) The latest innovation unveiled by Uber Kenya is their rider confirmation short code number. This is a secret 4 digit number sent by Uber directly to a passenger's smartphone once the uber vehicle arrives at the passenger's pick up point. Only after the passenger enters the uber vehicle and reveals this short code number to the driver can the journey to their final destination commence. Use one relevant example to justify whether you support or don't support this emerging trend [3 marks]

c.) Define the term 3D printing [2 marks]

d) . Differentiate between who is a luddite and non-luddite by relating any three arguments of his that match the two definitions [5 marks]

e) Explain the difference between a processor's model and a processor's generation e.g. for Intel or AMD processors? [1 mark]

f) Whereas transitioning from 2G to 3G was as simple as switching SIM cards on your phone, this was not the case when the current 4th generation was rolled out and most certainly will not be the case for its future successor the 5th generation of data communication standards. Provide a breakdown of groundwork for 4G implementation and two additional capabilities for 5G implementation [3 marks]

QUESTION THREE (20 Marks)

a) In the current 4th industrial revolution taking place right now across the developed world, technologies are merging with humans' physical lives. Describe how this revolution is gradually taking shape in the area of home automation by giving an example of how home automation works using three common smart home standards. Explain how one of the standards can be used to control your smart home from both within and outside [5 marks]

b) Whereas an individual computer can be locked down to prevent public use by pressing **Ctrl + Alt + Del** and selecting **lock this computer**, there are automated ways to perform device lock down in a multi-user computing environment. Explain how you would implement the principle of least privilege during device lockdown [3 marks]

c) Compare and contrast between traditional (SDLC) and Google Sheet LCD [2 marks]

d) Google maps on iphone and Android now shows drivers the location of police speed traps. Drivers share police trap positions and other updates to maps. The info is available to everyone navigating with Google. The feature comes from the Waze app which Google acquired in 2013. Make a reasoned argument either in support or against whether this emerging trend will improve transport in Kenyan society. [3 marks]

- e) 5G has been years in the making because the superior connectivity it promises to offer will transform everything. Defend the reasons why despite using higher frequency waves it has shorter wavelength [3 marks]
- f) Digital scent technology is an emerging trend which is just beginning. It allows transmission of smells over the internet. Draw a basic system design diagram illustrating the actors, as well as front and backend for such an application showing how it would work [4 marks]

QUESTION FOUR (20 Marks)

- a) In response to William “Bill” Joy’s widely published essay “why the future does not need us” Dr. John Messerley critically responded to Joy’s 9 point arguments about the consequences of rapidly expanding technology.. Analyse Messerley’s specific critique of Joy’s argument about “the notion of unintended consequences brought about by the emergence of 21st century technologies”. In your analysis highlight one relevant example in Kenya today which supports your answer [5 marks]
- b) I.T. asset management software does more than just create a list. Demonstrate how I.T. managers can use it to determine total cost of ownership (TCO) [3 Marks]
- c) Explain Jakob Nielsen Law [3 marks]
- d) “the hospital of the future is in the home” this statement by Sir Bruce Keogh was made in reference to the role the 4th industrial revolution will play in providing technologically advanced healthcare. Support this statement by justifying how analytics and computing will improve healthcare diagnostics, payment and information sharing [3 marks]
- e) Clothing technology mobile apps are an emerging trend just beginning to be used in Kenya for common fashion and style related tasks. You have just begun using a new belt app for your trousers. The app’s key feature behind its smart clothing tech service is running algorithms that measure the size of your waist and automatically adjust the belt on your trousers to comfortably fit your waistline . Draw a basic system design flow chart modelling the components of how the app works [4 marks]
- f) Define DNA digital storage [2 marks]

QUESTION FIVE (20 Marks)

- a) Explain what is meant by the term “A.I. snake oil” [2 marks]
- b) Use the current library and book lending and return process to analyse what kind of system Technical University of Kenya can make to meet the criteria of an emerging trend [5 marks]
- c) In his analysis of the characteristics of an emerging trend, the author Rotolo describes a “post-emergence period” which is a phase every technology experiences in it’s life where it begins to experience changes in the levels of the attributes or characteristics which defined it as an emerging trend. With reference to the definition of emerging trend, use the example of the latest social media app TikTok to determine what this post-emergence phase might look like for TikTok [5 marks]
- d) Explain how 4G LTE (Long Term Evolution) is different from 4G [3 marks]
- e) The 4th industrial revolution for finance will mean more than just using technology to pay for goods and services, or to send and collect money. Describe how fintech will disrupt the traditional banking business model and increase financial inclusion [3 marks]
- f) Explain how helium filled hard drives work [2 marks]